

Signals in scent

The living world is full of chemical signals, which we rarely notice. Produced by animals, many are dedicated scent chemicals called pheromones, which work like invisible control systems, affecting the way their own kind develop or behave. There are many types of pheromone, and they are often produced in microscopic amounts. Even so, their effects can be dramatic. Pheromones are used to mark territory, ward off danger, indicate when a female is ready to breed, and help a mother identify her young. They are particularly important for insects and other invertebrates, and for mammals with a better sense of smell than our own.



Male plains zebra



Calling card

While its baby clings to its back, a female ring-tailed lemur leaves a scent mark from her anal gland, just beneath her tail. Scent marking is very common in mammals. Hoofed mammals have scent glands on their feet, and others near their eyes, which they use for marking grass or twigs. By leaving their scent, mammals show others of their kind when they last passed by. Males use scent to claim territories and to warn rivals to keep away.

Potty trained

Domestic cats are careful to bury their droppings, but many other mammals leave theirs in conspicuous heaps, known as latrines. This European river otter is depositing its droppings, or spraint, to mark its territory. A male's territory can extend over 6 miles (10 km) of riverbanks or coasts. Inside its boundaries, the otter maintains dozens of latrine sites, showing other males that they are intruding on private ground.



Otter droppings mark its territory

Female Thomson's gazelle sniffs her newborn calf

